# Distance on the Coordinate Plane

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 6 — Relationships in Triangles |
| Lesson | Lesson 5 — Distance on the Coordinate Plane | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.8.GR.1.2 |

## 2. Standards

**MA.8.GR.1.2** Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

**Must.** Find the distance between two points: on one line by subtracting (across an axis too), slanted by the right triangle whose legs are read from the coordinates; an irrational answer exact and, when asked, to the tenth.

**Must not.** The distance formula (Clarification 2); radicands over 225; perimeters and maps with several legs (6.06).

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.2.1 | Demonstrate understanding by representing problems in multiple ways. | Notes II–III: one distance as a slanted segment, as a right triangle's legs and as $a^2 + b^2 = c^2$ |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | boards 3 and 6 split; the sixty-second re-vote |
| MA.K12.MTR.6.1 | Assess the reasonableness of solutions. | Notes I and board 9: a distance across zero checked by counting the squares |

## 4. Learning Target and Essential Question

**Learning target.** I can find the distance between two points: subtract on a line, and draw the right triangle when the segment is slanted.

**Essential question.** How can two pairs of numbers tell you how far apart two points are?

**Building on.** 6.03–04 — the Pythagorean Theorem for a missing side; Grade 6 (MA.6.GR.1.2) — the distance between points with the same x or the same y; Grade 7 — subtracting integers.

**Working toward.** 6.06 — perimeters and maps on the grid; Algebra 1 and Geometry — the distance formula, which is this triangle written as one line.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 11 | Teach from the front. Expand once per law; say the OFF line out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 3 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 11 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; finish tonight. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| Why is the distance from (−4, 3) to (5, 3) not 5 − 4? | DOK 2 |
| Two points have different x's and different y's. Why can you not just subtract? | DOK 3 |
| Does it matter which point you start from? Why does the answer come out the same? | DOK 2 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | How far apart are $(-7, -4)$ and $(-7, 6)$ ? | ***10 units*** | 2 — subtracted 6 − 4 across zero [Source of Truth, GR.1.2: integer errors across an axis] |
| 2 | How far apart are $(-9, 2)$ and $(3, 2)$ ? | ***12 units*** | 6 — subtracted 9 − 3 across zero [Source of Truth, GR.1.2: integer errors across an axis] |
| 3 | What is the distance between (−5, 1) and (3, 1)? | ***B — 8 units*** | (A) counted the grid lines, not the spaces [Math Nation names counting gridlines]; (C) subtracted 5 − 3 across zero [Source of Truth, GR.1.2: (−2, 0) to (3, 0) is not 1]; (D) −5 − 3: a distance is never negative [Notes I] |
| 4 | How far apart are $A(-4, -3)$ and $B(5, 9)$ ? | ***15 units*** | 21 — added the legs [MA.8.GR.1.2: the theorem, not a sum]; 'cannot, no triangle' — draw one [Math Nation names 'no triangle drawn'] |
| 5 | How far apart are $C(-2, -5)$ and $D(3, 3)$ , to the nearest tenth? | ***$\sqrt{89} \approx 9.4$ units*** | 13 — added the legs [MA.8.GR.1.2]; $\sqrt{(1^2 + 8^2)}$ — subtracted 3 − 2 across zero [Source of Truth, GR.1.2] |
| 6 | Which gives the distance between (4, 5) and (−3, −1)? | ***C — $\sqrt{(7^2 + 6^2)} = \sqrt{85}$*** | (A) 4 − 3 = 1 for the across leg — the sign of −3 dropped [Source of Truth, GR.1.2: integer errors across an axis]; (B) one point's coordinates taken as the legs [Math Nation names wrong horizontal and vertical lengths]; (D) added the legs [MA.8.GR.1.2: the theorem, not a sum] |
| 7 | How far is $(5, -12)$ from the origin? | ***13 units*** | 17 — added the legs [MA.8.GR.1.2]; 'the square root of −119' — subtracted 25 − 144 [the theorem adds the legs' squares] |
| 8 | On a town map in miles, the school is at $(-3, 2)$ , the library at $(4, 2)$ and the town hall at $(0, 5)$ . How far is the school from the library? | ***7 mi*** | 1 — subtracted 4 − 3 across zero [Source of Truth, GR.1.2]; a distance to the town hall — the wrong pair of points |
| 9 | Ty says (−3, 0) and (7, 0) are 4 units apart. What did he do? What is the distance? | ***He subtracted 7 − 3. Across zero the pieces add: 10 units.*** | '4 is right' — the axis-crossing error [Source of Truth, GR.1.2] |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** A printed grid for every board. The dashed legs are drawn on Notes II–III and the examples; from board 4 the student draws them. "Across, then up or down" said aloud with every slanted board.

**ELL.** **Horizontal**: side to side, like the horizon. **Vertical**: up and down. **Distance**: how far apart — never negative. Point at the grid with each word.

**Enrichment.** Find every point on the grid exactly 5 units from the origin with whole-number coordinates. (There are 12: (±5, 0), (0, ±5), (±3, ±4), (±4, ±3).)

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: Ty subtracted 7 − 3, but −3 and 7 are on opposite sides of zero, so the two pieces add: 3 + 7 = 10. A board that says only '10' without naming Ty's step has the answer and not the error.

**Vocabulary.** **horizontal distance**; **vertical distance**; **legs on the grid**

**Materials.** Whiteboards and markers; a printed grid for any student whose board has none. Calculators allowed, printed nowhere.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Find the distance between two points on a horizontal or vertical line — L9V; Find the distance between two points — ZBP.